Methodology of Complex Steganalysis Using Image Quality Characteristics Estimation

Yaroslav L. Grachev
chair of Control and Information Security
Federal State Institution
of Higher Education «Russian University of Transport»
Moscow, Russia
yaroslav446@mail.ru

Valentina G. Sidorenko
chair of Control and Information Security
Federal State Institution
of Higher Education «Russian University of Transport»
Moscow, Russia
valenfalk@mail.ru
ORCID: 0000-0003-4941-9008

***Abstract*—The paper presents a methodology for complex steganalysis that takes into account the quality characteristics of the analyzed image. The proposed method improves the accuracy of the joint application of statistical steganalysis methods, mainly by reducing false-positive results.**

***Keywords*— steganalysis, steganography, quality characteristics, binary classifier.**

I. Introduction

The threat of attackers using steganographic embedding techniques in image files to deliver malicious data inside attacked systems is growing in urgency [1]. The greatest risks in this context relate to systems related to the processing and storage of graphic information - for example, vision systems, monitoring and visualization systems for autonomous and unmanned vehicles, corporate systems for processing and storing documents.

The typical scenario of steganography being used by attackers is its use as part of the implementation of malicious loaders such as IDAT Loader, Zero.T and their like [1]. They are the auxiliary means that ensures «infection» - invisible penetration of malicious code inside the attacked system and ensuring its ability to launch. Thus, the use of steganography and the organization of a glass channel is part of the activities to organize a reliable and safe from the point of view of attacker's way of transmitting malicious information.

The most frequent method of steganographic embedding of data into images used by attackers as part of cyberattacks is the method of hiding pixel color intensity (luminance) estimations in the Least Significant Bits (LSB) [1].

II. Methodology of Complex Steganalysis

Currently, there are two main approaches to the development of steganalysis methods [2]:

1. The use of artificial convolutional neural networks (CNN).

2. The use of binary classifiers, including those based on extensive feature spaces and the ensemble approach.

The first approach is more actively developed today, but it does not always guarantee higher accuracy of steganography detection with more complex computational implementation of the steganalytical system (in particular, in the issues of training and retraining of the neural network) and less interpretability of the results [2].

This paper proposes a method of complex steganalysis of embedding in LSB based on the second approach: selecting a set of meaningful features and training a binary classifier to solve the problem of distinguishing between filled and empty stego-containers. Complex steganalysis includes the results of statistical steganalysis methods as well as additional computable or intrinsic image features and parameters.

Based on the chosen approach, it is possible to formalize the methodology of complex steganalysis in the form of an algorithm for implementing the steganalysis method:

1. The choice of statistical methods of steganalysis that form a numerical estimation (estimates of the relative amount of hidden information or another estimation that determines the estimated probability of hidden information) or a binary detection estimate (whether hidden information is present in the image) and the calculation using the selected methods.

2. The choice for a set of additional characteristics and attributes:

- intrinsic to the
- calculated on the basis of image data (e.g., various statistical characteristics of the image not directly related to steganalysis tasks).

Calculation of the estimations of the selected computed features.

3. Classification of the stego-container of the analyzed image using a pre-trained binary classifier based on a set of features including all estimations from steps 1 and

III. Selection of Features for Complex Steganalysis

The paper [3] demonstrated the existence of statically significant correlation between the magnitude of error in estimating the relative amount of hidden information for empty containers by Chi-Square Attack (CSA) [4,5] and Regular-Singular (RS) [6] methods and a number of absolute quantitative estimates of qualitative image characteristics: noise, sharpness, blurriness, contrast, Shannon and Renyi entropy.

The choice of the above-mentioned characteristics is conditioned by the hypothesis formed in the course of solving practical problems on detection of steganographic embedding in graphic files about deterioration of steganalysis accuracy for images that can be characterized as "low-quality" during visual inspection. The work [3] demonstrates, using several examples, how a noticeable visual deterioration of image quality correspondingly affects the quality scores generated by the chosen algorithms for their calculation: the scores by CSA and RS methods turn out to be noticeably higher for, for example, "blurred", "grainy" and overexposed images.

To further confirm the influence of visually perceived image quality on the accuracy of steganalysis methods, a study of the relationship between quality scores and errors in estimating the relative amount of hidden information by CSA and RS methods for filled containers was conducted. A sample of 16,614 PNG-24 files was prepared. For all images, qualitative characteristics were calculated. After that, three subsamples were formed:

- PERFECT (images with low noise and blur scores, high sharpness and entropy scores),
- NORMAL (images with low or medium noise and blur scores, medium or high sharpness and entropy scores),
- BAD (images for which at least two low quality conditions are met: high noise, high blur, low sharpness, low entropy).

Contrast estimation was not taken into account, as there is no obvious or visually intuitive criterion for assigning it to the fuzzy estimation of "bad" or "good" [7]. The k-means clustering method (k-means++) is used to perform fuzzy classification of grade levels (high, medium, low) [8].

For all images of each of the three samples:

- the mean errors in the estimation of the relative volume of hidden information by CSA and RS methods with an empty container were calculated;
- hiding of information of different volumes in LSB was performed first sequentially, then pseudo-randomly;
- average estimations of errors in estimating the relative volume of hidden information by CSA and RS methods for both variants of filled containers were calculated;
- calculated mean estimations of errors in estimating of CSA and RS for both variants of filled containers for each range of relative volume of really embedded information: [0%-10%, ..., 90%-100%].

To determine the influence of image quality on the results of steganalysis methods, let us consider the obtained errors in estimation in determining the relative amount of hidden information: they are presented in Figures 1-4.

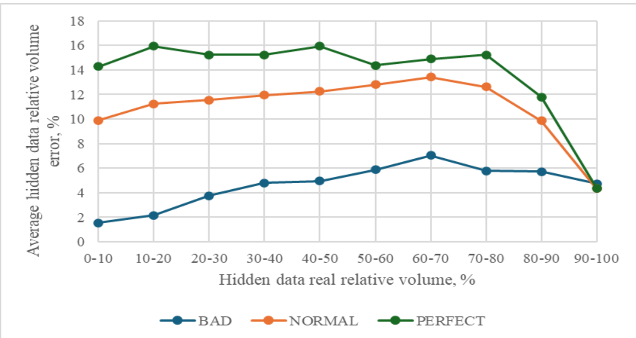

Fig. 1. Comparing the mean deviations of the CSA method for sequential embedding

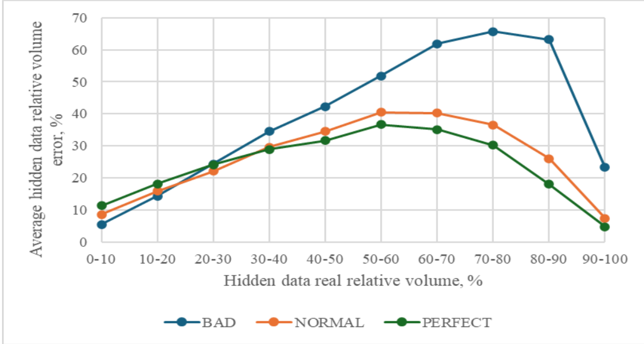

Fig. 2. Comparing the mean deviations of the CSA method for pseudorandom embedding

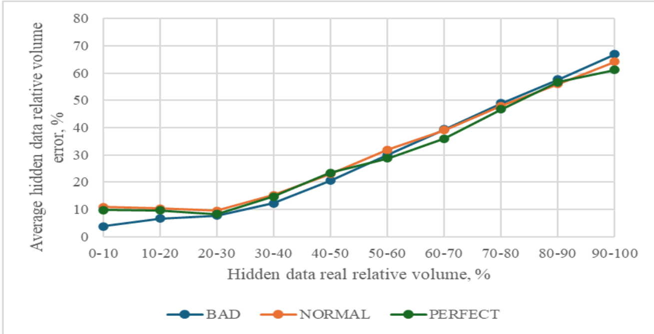

Fig. 3. Comparing the mean deviations of the RS method for sequential embedding

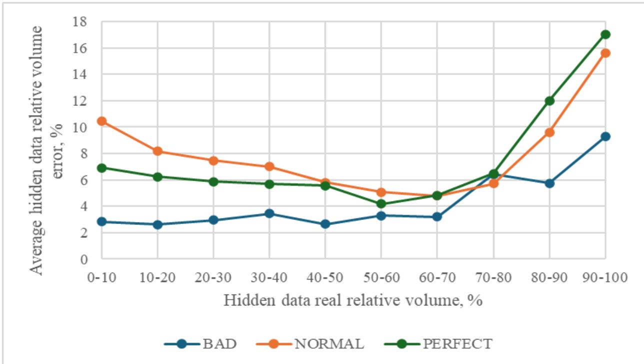

Fig. 4. Comparison of mean deviations of the RS method in pseudorandom embedding

On the basis of the results obtained, it can be concluded that image quality has an impact on the accuracy of steganalysis by CSA and RS methods for filled containers (except for the analysis of the container with sequential embedding by RS method).

The discovered dependence is also confirmed by the results of Spearman's correlations calculation (Student's t-criterion [9] is used to calculate statistical significance, Cheddock's scale [10] is used to estimate the correlation strength) between the error estimation in estimating the relative amount of hidden information by the corresponding steganalysis methods and the estimations of the considered absolute qualitative characteristics of the image (Tables 1-4).

TABLE I. CALCULATING OF THE CORRELATION BETWEEN THE ERROR OF THE CSA METHOD AND ESTIMATES OF QUALITATIVE CHARACTERISTICS IN SEQUENTIAL EMBEDDING

	Correlation coefficient	t-criterion estimation	Is the correlation significant?	Degree
Noise	0,295	34,91	yes	low
Sharpness	0,294	34,85	yes	low
Blur	-0,453	-57,46	yes	moderate

	Correlation coefficient	t-criterion estimation	Is the correlation significant?	Degree
Contrast	-0,053	-6,05	yes	low
Shannon	0,17	19,56	yes	low
Renyi	0,173	19,89	yes	low

TABLE II. CALCULATING THE CORRELATION BETWEEN THE ERROR OF THE CSA METHOD AND ESTIMATES OF QUALITY CHARACTERISTICS IN PSEUDORANDOM EMBEDDING

	Correlation coefficient	t-criterion estimation	Is the correlation significant?	Degree
Noise	-0,131	-14,93	yes	low
Sharpness	-0,107	-12,20	yes	low
Blur	0,15	17,21	yes	low
Contrast	-0,023	-2,59	yes	low
Shannon	-0,1	-11,42	yes	low
Renyi	-0,102	-11,54	yes	low

TABLE III. CALCULATING THE CORRELATION BETWEEN THE ERROR OF THE RS METHOD AND ESTIMATES OF QUALITY CHARACTERISTICS IN SEQUENTIAL EMBEDDING

	Correlation coefficient	t-criterion estimation	Is the correlation significant?	Degree
Noise	0,078	8,901	yes	low
Sharpness	0,067	7,452	yes	low
Blur	-0,091	-10,288	yes	low
Contrast	-0,029	-3,375	yes	low
Shannon	0,008	0,949	no	—
Renyi	0,008	0,932	no	—

TABLE IV. CALCULATING THE CORRELATION BETWEEN THE ERROR OF THE RS METHOD AND ESTIMATES OF QUALITY CHARACTERISTICS IN PSEUDORANDOM EMBEDDING

	Correlation coefficient	t-criterion estimation	Is the correlation significant?	Degree
Noise	0,318	37,96	yes	moderate
Sharpness	0,349	42,10	yes	moderate
Blur	-0,467	-59,72	yes	moderate
Contrast	0,026	2,89	yes	low
Shannon	0,173	19,87	yes	low
Renyi	0,169	19,39	yes	low

Beside absolute estimations of image quality characteristics, it is suggested to take into account the sensitivity of RS method to statistical anomalies in the frequency representation of the image (Table 5). Since the key task is the detection of embedding in the LSB, possible embedding or any other artificial distortions in the coefficients of the DCT matrices should be considered as additional disturbances affecting the accuracy of steganalysis by the RS method. In this regard, it is proposed to include in the resulting set of features the results of a simple steganalysis of possible embedding performed by the Koch-Zhao method in sequence.

TABLE V. EXAMPLES OF CHANGES IN RS METHOD SCORES WHEN HIDING DATA USING THE KOCH-ZHAO METHOD FOR A TEST IMAGE

Volume of information hidden by the Koch-Zhao method, bits	Estimation of relative volume of hidden information by RS method, %
0	3,46
426	4,03
822	5,16
2 484	9,66

As the results of steganalysis by the algorithm described in [11], it is proposed to select the estimated bit volume of embedded information and the estimated value of the embedding threshold for the generated feature set. Using both estimations will reduce the probability of false detection of embedding in the frequency domain. To signal the detection of such embedding, it is necessary to obtain both a relatively high estimated threshold value and a significant estimated bit-length of the hidden data.

Another significant factor is the direction of traversal of the pixel matrix (or pixel blocks) during image embedding and steganalysis. Embedding in the LSB in the sequential version can be performed either by rows or columns of the pixel matrix. In the second case, the steganalysis accuracy of the CSA method, which traditionally uses row-by-row traversal, will be extremely low. In particular, when testing the previously described NORMAL subsample (and its variants, in which information embedding into the LSB was performed sequentially and pseudo randomly), the following average error values in estimating the relative amount of embedded information were obtained:

- 31.67% when hiding by columns and steganalysis by rows;
- 30.55% when hiding by rows and steganalysis by columns.

The similar errors in estimation for steganalysis by the above-mentioned sequential hiding detection algorithm implemented by the Koch-Zhao method are 48.66% and 49.34%, respectively.

Thus, in order to realize an effective comprehensive steganalysis, it is necessary to perform analytical algorithms, whose accuracy potentially depends on the way the pixel matrix is traversed sequentially, twice: row-by-row and column-by-column, and to consider both sets of results within the set of features under consideration.

Due to the fact that the basis of the proposed method of complex steganalysis is statistical methods and statistical characteristics, the accuracy of the obtained results can be affected by the volume of analyzed data - that is, in this case, the number of image pixels (and thus, in fact, its linear dimensions).

To confirm this assumption, the above-mentioned NORMAL subsample was used in three variants (without embedding, with sequential and pseudo-random embedding in the LSB). The correlation between the number of image pixels and the error in estimating the relative amount of hidden information by CSA and RS methods was calculated (Tables 6-8).

TABLE VI. CORRELATION COEFFICIENTS BETWEEN THE ERRORS OF CSA AND RS METHODS AND THE TOTAL NUMBER OF IMAGE PIXELS

Embedding	CSA error	RS error
No embedding	-0,498	-0,488
Sequential	-0,597	-0,086
Pseudorandom	0,137	-0,334

TABLE VII. THE STUDENT'S T-TEST ESTIMATIONS OF THE CORRELATION COEFFICIENTS BETWEEN THE ERRORS OF THE CSA AND RS METHODS AND THE NUMBER OF PIXELS

Embedding	CSA error	RS error
No embedding	-73,97	-71,97
Sequential	-79,17	-9,20
Pseudorandom	14,65	-37,71

TABLE VIII. THE DEGREE OF SIGNIFICANCE FOR THE CORRELATION BETWEEN THE ERRORS OF THE CSA AND RS METHODS AND THE NUMBER OF IMAGE PIXELS

Embedding	CSA error	RS error
No embedding	Moderate	Moderate
Sequential	Noticeable	Low
Pseudorandom	Low	Moderate

Thus, the number of pixels as an inherent characteristic of an image should be included in the feature set of a comprehensive steganalysis that combines statistical methods and features.

IV. BINARY COMPLEX STEGANALYSIS CLASSIFIER

The set of features taken into account by the proposed method of complex steganalysis includes:

- estimation of the amount of hidden information by the CSA method (horizontal traversal);
- estimation of hidden information volume by CSA method (vertical traversal)
- estimation of the amount of hidden information by the RS method;
- estimation of noise level;
- estimation of sharpness level;
- estimation of blur level;
- contrast level estimation;
- Shannon entropy.
- Renyi entropy;
- estimated concealment threshold according to the evaluation of the steganalysis algorithm of concealment performed by the Koch-Zhao method (horizontal traversal);
- the estimated bit length of the hidden data according to the Koch-Zhao steganalysis algorithm (horizontal traversal);
- estimated hidden data threshold according to the evaluation of the steganalysis algorithm of hiding performed by the Koch-Zhao method (vertical traversal);
- estimated bit length of the hidden data according to the evaluation of the steganalysis algorithm of hiding performed by the Koch-Zhao method (vertical traversal);
- number of image pixels.

In order to train the binary classifier, a sample of 29,856 image files of different size and quality was prepared. In 45% of the files the information of random size was embedded randomly (pseudo-randomly or sequentially - by rows or columns), in 10% random data were embedded by the Koch-Zhao method, in the remaining 45% no embedding was performed. To form the sample, we used datasets of images with pre-implemented embedding in LSB, high-resolution, natural and artificial, with initially different indices of qualitative characteristics: [12-16].

Training of a binary classifier using the ML.NET framework was performed [17]. 20% of randomly selected sample data was used as a validation dataset. The highest accuracy during training, 0.9282, was demonstrated by the random forest model trained with the following hyperparameters [18]:

- number of trees: 137;
- maximum number of "leaves" of the tree: 356;
- percentage of attributes used in the construction of the tree: 80%.

Another non-overlapping sample was prepared for testing the model, which was compiled according to the same principles as the training sample. It included 2,642 images. The quality metrics of the tested model are presented in Table 9, Table 10, Figures 5 and 6 show the corresponding error matrix, ROC and PR curves.

TABLE IX. METRICS FOR THE QUALITY OF THE TRAINED MODEL

Metric	Estimation
Accuracy	0,947
Precision	0,959
Recall	0,941
F1 Score	0,95
AUC	0,985
AUCPR	0,989

TABLE X. ERROR MATRIX OF THE TRAINED MODEL

		Model estimation	
		Hiding is defined	Hiding is not defined
Real data	Hidden	1330	84
	Unhidden	56	1172

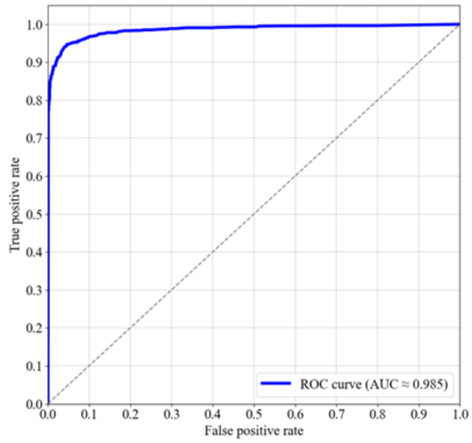

Fig. 5. ROC-curve

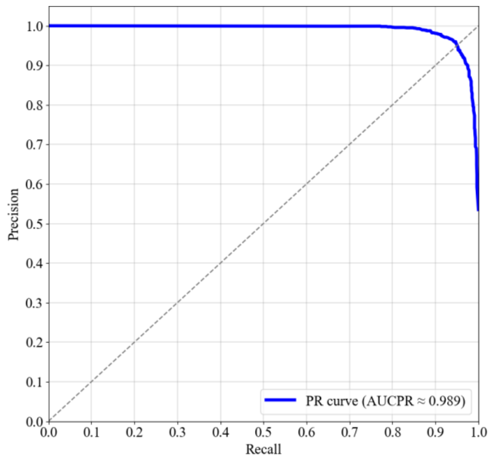

Fig. 6. Precision-Recall curve

In order to assess the significance of including selected additional features in the feature set for training, a number of additional random forest models were trained on the same dataset but with a truncated feature set. Their quality metrics are summarised in Table 11.

- (1): without quality characteristics.
- (2): without vertical traversals.
- (3): without considering the number of pixels.
- (4): without Koch-Zhao embedding steganalysis estimates.
- (5): only CSA and RS estimates.

TABLE XI. METRICS OF MODEL QUALITY WHEN TESTED ON A SAMPLE WITH EMBEDDING IN 2 LSB

	(1)	(2)	(3)	(4)	(5)
Accuracy	-0,04	-0,005	-0,014	-0,044	-0,057
Precision	-0,026	0,002	-0,023	-0,013	-0,08
Recall	-0,05	-0,013	-0,002	-0,075	-0,018
F1 Score	-0,039	-0,019	-0,013	-0,045	-0,049
AUC	-0,018	-0,06	-0,005	-0,02	-0,03
AUCPR	-0,015	-0,062	-0,005	-0,015	-0,027

The results confirm the significance of including the selected additional characteristics in the set of analyzed features within the framework of complex steganalysis.

Since, on average, the strength of correlation between quality scores and the error in estimating the relative amount of embedded information using CSA and RS methods is higher when analyzing empty containers than filled ones, taking into account these characteristics allows us to reduce the probability of false positive detection of steganography associated with poor image quality.

The trained model was tested on a sample composed similarly to the previous ones, when embedded in two LSBs instead of one. The quality metrics and error matrix of the tested model are presented in Tables 12-13.

TABLE XII. METRICS OF MODEL QUALITY WHEN TESTED ON A SAMPLE WITH EMBEDDING IN 2 LSB

Metric	Estimation
Accuracy	0,922
Precision	0,96
Recall	0,889
F1 Score	0,923
AUC	0,981
AUCPR	0,984

TABLE XIII. ERROR MATRIX OF THE MODEL WHEN TESTED ON A SAMPLE WITH EMBEDDING IN 2 LSB

		Model estimation	
		Hiding is defined	Hiding is not defined
Real data	Hidden	1 243	155
	Unhidden	57	1 187

In this case, the proportion of false positives was 57 images - about 2.16%. This is one image more than when the model was tested on a sample in which the embedding was performed in a single NCB. Thus, the model that takes into account the selected qualitative and additional characteristics is able to effectively resist false detection of low-quality images as containing embedded data.

V. CONCLUSION

The proposed method of complex steganalysis allows to form a conclusion about the presence or absence of steganographic embedding in the LSB of graphic files taking into account possible distortions in the estimates of statistical methods of steganalysis when working with low-quality images.

This method can be integrated into another ensemble steganalysis method as a component part or used to implement an independent separate steganalytical detector.

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